Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years of experience building **Python/C++ services**, **distributed orchestration**, and reliable **robotics-adjacent infrastructure** across AWS and Kubernetes.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **fault-tolerant orchestration** with explicit job-state transitions, failure isolation, recoverable background processing, and observable concurrent workflows under production load reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **production backend pipelines**, implementing incremental processing, validation, reconciliation, error handling, cloud storage, repeatable CI/CD-backed releases, and stable production observability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive customer workflows during concurrent model execution and production traffic reliably.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **maintainable microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments at scale.
- Increased pipeline efficiency by 68% through **AWS Glue and S3 automation**, transforming structured data while reducing storage costs by 15% and supporting repeatable cloud delivery workflows at scale.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **microservice platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with Node.js, WebSockets, and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **real-time state synchronization**, integrating automated tests, Docker/Jenkins deployments, Azure Monitor, Sentry, and Kibana across modular production services and delivery workflows reliably.

Skills

Languages: Python, C++, Java, SQL, JavaScript, TypeScript, Bash
Backend & Systems: Distributed Systems, Microservices, FastAPI, Flask, REST APIs, API Design, Data Modeling, Multithreading, State Machines
Cloud & Platform: AWS, Azure, GCP, Kubernetes, Docker, AWS RDS, AWS S3, AWS Glue
Reliability & Operations: Fault Isolation, Observability, Incident Diagnostics, Performance Tuning, Data Validation, Error Handling, Sentry, Azure Monitor
Engineering & AI: CI/CD, GitHub Actions, Jenkins, Git, PyTest, Mockito, Unit Testing, Integration Testing, Transformers, Reinforcement Learning

Projects

Robot Workflow Orchestrator - Design Concept

- Proposed a **Python orchestration service** coordinating long-running robot tasks through explicit state machines, idempotent commands, checkpointed recovery, priority queues, operator overrides, safe retries, and versioned robot-facing APIs.

Fleet Operations Observability - Design Concept

- Proposed a **Kubernetes-based operations platform** aggregating service and robot events, correlating workflow failures, exposing health dashboards, triggering alerts, preserving audit trails, and supporting incident diagnostics across distributed fleets.

CPU Scheduling Simulator | Live

- Implemented **C++ scheduling algorithms** for FCFS, SJF, SRT, Round Robin, priority, and MLQ, with edge-case handling, process-state visualization, and dynamic turnaround and waiting-time calculations across simulations reliably.